

John "J.D." Squire

506 S Mathews Ave, Urbana, IL 61801

(210) 649-7458 | jdsg2018@gmail.com | jsquire4@illinois.edu | <https://john-squire.github.io>

I am a fourth-year medical student with a passion for engineering and medicine. I am currently pursuing a residency position in diagnostic radiology. By harnessing the power of technology today and of the future, I believe we can improve the quality of life for everyone around the globe.

Education

Doctor of Medicine	May 2026
Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign, Urbana, IL	
Bachelor of Science in Biomedical Engineering, <i>summa cum laude</i>	May 2022
The University of Texas at Dallas, Richardson, TX	

Ongoing Research Projects

Magnetic Resonance Functional Imaging Laboratory (MRFIL), University of Illinois at Urbana-Champaign

- My research within MRFIL centers on applying machine learning techniques to magnetic resonance elastography (MRE), with the goal of identifying and masking brain structures in an image. MRE holds promise for earlier detection of degenerative neurological diseases, but its widespread use is limited by the time-intensive image processing required. My work aims to develop AI tools that streamline this processing, potentially accelerating research and clinical workflows.

Data Science Project: A Comparison of Machine Learning Models in the Detection of Pulmonary Embolism

- According to the Radiological Society of North America, it is time consuming to diagnose a pulmonary embolism and is susceptible to overdiagnosis. My data science project focuses on comparing classification and segmentation machine learning models in the identification of pulmonary embolism in computed tomography pulmonary angiography images, including developing my own segmentation model, evaluating the performance of binary classification versus image masking in providing useful information to the radiologist.

Capstone Project: Apollo Health

- Current electronic medical record (EMR) systems are fragmented and inefficient, especially in the storage and transfer of health data, presenting a critical barrier to delivering high-quality, coordinated healthcare. I am a part of a team of medical and business students developing Apollo Health, an EMR platform that integrates blockchain-based data ownership, AI-enabled clinical decision support, and a modern user-centric design. By integrating AI and blockchain, we encourage patient ownership of data, personalized health insights with future wearable integration, and easy provider-to-provider data transactions, even if they are a part of different health systems.

Research Experience

Graduate Researcher	Dec 2022 – present
<i>Magnetic Resonance Functional Imaging Laboratory, University of Illinois at Urbana-Champaign</i>	
Developed machine learning methods to extract information about the brain in magnetic resonance elastography data to facilitate clinical and research elastography applications	
Graduate Researcher	Dec 2022 – Dec 2024
<i>Mobility and Fall Prevention Research Laboratory, University of Illinois at Urbana-Champaign</i>	
Characterized cognitive and motor function in gait with the goal of developing objective and standardized diagnostic measures of motor impairment and treatment	
Undergraduate Researcher	Aug 2021 – Jun 2022
<i>Molecular Imaging and Optical Nanotherapeutics Laboratory, University of Texas at Dallas</i>	
Characterized various nanoscale drugs for head and neck cancers under light-activated, x-ray activated, and γ-ray activated conditions	
Premier College Intern	May – Aug 2021
<i>Air Force Research Laboratory, Wright-Patterson Air Force Base</i>	
- Worked in the 711th Human Performance Wing in the Bioanalytics focusing on toxicology research - Utilized HTML, CSS, JS, PHP, Python, GUI development, and machine learning	
Undergraduate Researcher	Jan – May 2021
<i>Molecular Radiation Biology Research Laboratory, University of Texas Southwestern Medical Center</i>	
- A research partnership between UT Dallas and UT Southwestern under the 2021 Green Fellowship, a selective program that provided a stipend to work at a UT Southwestern research laboratory for a semester - Acquired imaging data in 3D tumor models for head and neck cancer as well as imaged and modeled time-dependent distribution of x-ray responsive nanoparticles in response to varying doses of radiation therapy	

John “J.D.” Squire

506 S Mathews Ave, Urbana, IL 61801

(210) 649-7458 | jdsg2018@gmail.com | jsquire4@illinois.edu | <https://john-squire.github.io>

- Investigated the use of light-activatable nanotechnology as a facilitator for radiation therapy in radiation-resistant tumors of the head and neck

Undergraduate Researcher

Jan – Dec
2020

Systems for Augmenting Human Mechanics Laboratory, University of Texas at Dallas

- Learned about neural network architecture and building a neural network to identify diseases related to the hip joint

Undergraduate Researcher

Jun 2018 –
Jun 2019

Speech Disorders and Technology Laboratory, University of Texas at Dallas

- Clark Summer Research Program participant, a program that provided on-campus housing and a stipend to work in an on-campus research laboratory
- Developed MATLAB programs that connected speech data with machine learning model

Publications

Finneran M, **Squire J**, Gordhan A, Nardone E. Dural arteriovenous fistula mimicking a stroke: A misdiagnosis of two months. *Radiology Case Reports*. 2024;19(12):5995-5999. <https://doi.org/10.1016/j.radcr.2024.09.043>

Squire JD, Anderson AT, Johnson CL, Sutton BP. ANTs, BET, or...neither? An exploration of brain masking and machine learning tools applied to magnetic resonance elastography. In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.; 2024:1-4. <https://doi.org/10.1109/EMBC53108.2024.10781675>

Shah N, **Squire J**, Guirguis M, Saha D, Hoyt K, Wang KK-H, Agarwal V, Obaid G. Deep-Tissue Activation of Photodynamic Medicines: An Update and Clinical Perspectives. *Cancers*. 2022; 14(8):2004. <https://doi.org/10.3390/cancers14082004>

Presentations

“The Intra-Fallopian Device: The novel reversible form of contraception without the regret of surgical sterilization,” 2nd Place Pitch Competition, Cozad New Venture Challenge, University of Illinois at Urbana-Champaign. Champaign, IL. April 2025.

“MREureka!: Efficiently and Effectively Segmenting Brain Elastographs with nnU-Net,” Finalist Presentation, Global Consortium of Innovation and Engineering in Medicine, Carle Illinois College of Medicine. Urbana, IL. April 2025.

“Arthritis to Meningitis: A rare case of rheumatoid meningitis,” Global Consortium of Innovation and Engineering in Medicine, Carle Illinois College of Medicine. Urbana, IL. April 2025.

“A student-faculty partnership to support preparation for USMLE Step 1,” Medical Education Experience Conference, Carle Illinois College of Medicine. Champaign, IL. January 2025.

“Radiologic Confirmation of Chilaiiditi Syndrome in an Adult Female,” Southern Radiological Conference, Southern Radiological Society. Point Clear, AL. January 2025.

“ANTs, BET, or...neither? An exploration of brain masking and machine learning tools applied to magnetic resonance elastography,” 2024 Annual International Conference of the IEEE Engineering in Medicine and Biology Society, IEEE Engineering in Medicine and Biology Society. Orlando, FL. July 2024.

“A Teaching Electronic Medical Records System for Preclinical Students towards Clinical Simulation Evaluation,” Medical Education Experience Conference, Carle Illinois College of Medicine. Champaign, IL. January 2024.

“Tumestent: A vascular solution for erectile dysfunction,” 1st Place Presentation, Penile Prosthesis Innovations, Carle Illinois College of Medicine. Urbana, IL. October 2023.

“Becoming a Military Clinician,” Carle Illinois College of Medicine. Champaign, IL. July 2023.

“Made in China 2025: A Health Perspective,” Officer Training School, U.S. Air Force. Maxwell AFB, AL. June 2023.

“Bonobo.care: Altruism Redefined,” Cozad New Venture Challenge, University of Illinois at Urbana-Champaign. Champaign, IL. April 2023.

“Efficacy of Nanoformulation-Enhanced Photodynamic Therapy Excited by Radiation in Head and Neck Cancer Cells,” Green Fellowship, UT Southwestern Graduate School of Biomedical Sciences. Dallas, TX. May 2021.

“Wavcutter: A Tool for Speech Articulation Data Preprocessing,” Anson L. Clark Summer Research Program, University of Texas at Dallas. Richardson, TX. August 2018.

Honors and Awards

Second Place Pitch, Cozad New Venture Challenge

May 2025

- Awarded \$500 for achieving second place for the Cozad New Venture Challenge Pitch Competition
- Pitched in front of investors about the Intra-Fallopian Device, a novel reversible long-term contraception start-up concept for the Health Innovation Track

John "J.D." Squire

506 S Mathews Ave, Urbana, IL 61801

(210) 649-7458 | jdsq2018@gmail.com | jsquire4@illinois.edu | <https://john-squire.github.io>

Finalist Presentation, Global Consortium of Innovation and Engineering in Medicine	Apr 2025
Invited to host an oral and poster presentation about my research in applying machine learning to magnetic resonance elastography in the brain	
Air Force Health Professions Scholarship Program	May 2022 – present
- Four-year scholarship to cover medical school tuition and fees - Serving in the Air Force Reserve as a 2nd Lieutenant	
Carle Illinois College of Medicine Scholarship	May 2022 – present
Four-year scholarship covering \$7,000 of educational expenses each year	
First Place for Website Design, FBLA-PBL National Leadership Conference	Jun 2020
- Awarded \$1,000 for achieving first place for the Website Design challenge - Achieved first place at the Texas State Leadership Conference to qualify for nationals	
San Antonio Pipeliners Association Scholarship	Jul 2019 – Jul 2020
- Recipient of the 2019 scholarship covering \$4,000 of educational expenses - Recipient of the 2020 scholarship covering \$5,000 of educational expenses	
First Place for Mobile Application Design, FBLA-PBL National Leadership Conference	Jun 2019
- Awarded \$1,000 for achieving first place for the Mobile Application Design challenge - Achieved first place at the Texas State Leadership Conference to qualify for nationals	
Society of American Military Engineers Scholarship	May 2019 – May 2020
- Recipient of the 2019 SAME San Antonio Post Scholarship covering \$5,000 of educational expenses - Recipient of the 2020 SAME San Antonio Post Scholarship covering \$3,000 of educational expenses - Recipient of the 2020 SAME Fort Worth Post Scholarship covering \$2,500 of educational expenses	
Academic Excellence Scholarship	Aug 2018 – May 2022
Four-year scholarship to cover undergraduate tuition and fees at the University of Texas at Dallas	
The First Tee Scholars Program	Aug 2018 – May 2022
Selected as the First Tee Scholar from the First Tee of Greater San Antonio program, receiving \$2,500 for educational expenses each year for four years	
Burger King Scholars Program Scholarship	Aug 2018
Recipient of the 2018 Burger King Scholars award covering \$1,000 of educational expenses	
National Honor Society Scholarship	Aug 2018
Selected as a National Semi-Finalist for the 2018 NHS Scholarship Program, receiving \$2,850 for educational expenses	

Skills and Proficiencies

- Basic clinical skills in family medicine, internal medicine, pediatrics, surgery, ob/gyn, neurology, and psychiatry through the Carle Illinois College of Medicine
- Entrepreneurship skills in healthcare innovations and customer discovery through the National Science Foundation I-Corps Program and the Cozad New Venture Challenge
- Research skills in cell culture, nanoparticle synthesis, and machine learning through various research projects
- Programming skills in Java, Python, HTML, CSS, PHP, JavaScript, MATLAB, Arduino, and Microsoft PowerApps through self-teaching and research projects
- Engineering skills in SolidWorks CAD and 3D printing through various engineering projects
- Video editing skill in VEGAS Pro 15 through personal projects and presentations

Professional Activities

Treasurer/Marketing Chair, Personal Finances and Investments Interest Group	Dec 2024 – present
<i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i>	
Served as a founding executive board members, encouraging students to become involved in managing their own finances and how to begin building their wealth beginning from medical school	
President, Federal Medical Professionals Interest Group	Sept 2023 – present
<i>Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign</i>	
Served as a founding executive board member in creating a supportive environment for other medical students who plan to serve in federal position after medical school, like HPSP students	

John “J.D.” Squire

506 S Mathews Ave, Urbana, IL 61801

(210) 649-7458 | jdsg2018@gmail.com | jsquire4@illinois.edu | <https://john-squire.github.io>

Marketing Chair, CIMED Student Government

Dec 2023 –

Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign

Dec 2024

- Served as a member of the voting body and the main point of contact for marketing affairs

Member, Radiology Interest Group

Jun 2022 –

Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign

present

- Engaged in meetings revolving around the practice of radiology

Vice President of Marketing, Texas Phi Beta Lambda/Future Business Leaders of America

Aug 2020 –

Naveen Jindal School of Management, University of Texas at Dallas

May 2022

- Served as an executive voting member and marketing liaison for the Texas region of PBL/FBLA chapters

Professional Memberships

- American College of Radiology (ACR)
- Association of Military Surgeons of the United States (AMSUS), The Society of Federal Health Professionals
- International Society for Magnetic Resonance in Medicine (ISMRM)
- IEEE Engineering in Medicine and Biology Society (IEEE EMBS)
- Illinois State Medical Society (ISMS)
- Society of American Military Engineers (SAME)
- National Society of Professional Engineers (NSPE)

Volunteer Activities

Voting Member, Clinical Curriculum Subcommittee

Sept 2023 –

Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign

present

- Served as voting member to bring a student perspective of the clinical curriculum and improve it for future classes

Voting Member, Electives Subcommittee

Sept 2023 –

Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign

present

- Served as a voting member to make decisions on proposed permanent and student created classroom, clinical, or research electives

Mentor, Wellness Mentoring Program

Jun 2023 –

Carle Illinois College of Medicine, University of Illinois at Urbana-Champaign

Jun 2024

- Volunteered as a mentor to a first-year medical student, easing their transition to medical school life and being an overall resource for academics and wellness

Mentor, Freshman Mentor Program

Aug 2019 –

Office of Undergraduate Education, University of Texas at Dallas

May 2022

- Volunteered as a mentor to freshman students transitioning to college life, providing mentorship in academic engagement, social involvement, and community service

Voting Member, Victor L. Worsfold Grant Program Fund Selection Committee

Aug 2018 –

Hobson Wildenthal Honors College, University of Texas at Dallas

May 2022

- Met with a committee each semester and selected student-led volunteer initiatives to be funded by the Worsfold Grant